

Comparing the Generation of Renewable and Nonrenewable Energy Overtime

1 INTRODUCTION

This report examines long-term trends in Canadian electricity generation from 2005 to 2016, comparing renewable and non-renewable energy sources. The goal of the analysis is to quantify how production levels have changed over time, evaluate differences in variability between the two categories, and determine whether meaningful shifts occurred across two distinct periods: 2005–2010 and 2011–2016. As stated in the original document, the study “provides a comprehensive view of how Canada’s energy landscape evolved during a decade marked by technological growth and increasing emphasis on sustainability” . Using statistical modeling, hypothesis testing, and visualization, the analysis highlights how Canada’s energy mix has shifted toward cleaner sources.

2 DATA DESCRIPTION & PREPARATION

The dataset used in this analysis was obtained from the Government of Canada Open Data Portal and contains annual electricity generation values measured in gigawatt-hours (GWh). Each record includes the energy source, region, and year. Only national-level data for Canada were included in this study.

To focus on national trends, the dataset was filtered to include only the Canada region. Energy sources were grouped into two categories:

- **Renewable:** Hydro, Wind, Biomass, Solar
- **Non-renewable:** Nuclear, Coal, Natural Gas, Oil and Diesel

After filtering the dataset to include only Canadian totals, the energy source variable was converted to a factor to support grouping operations. Annual totals were then aggregated for each category, resulting in two summary datasets: one containing yearly renewable generation

and another containing yearly non-renewable generation. These aggregated datasets formed the basis for all statistical modeling and visualization.

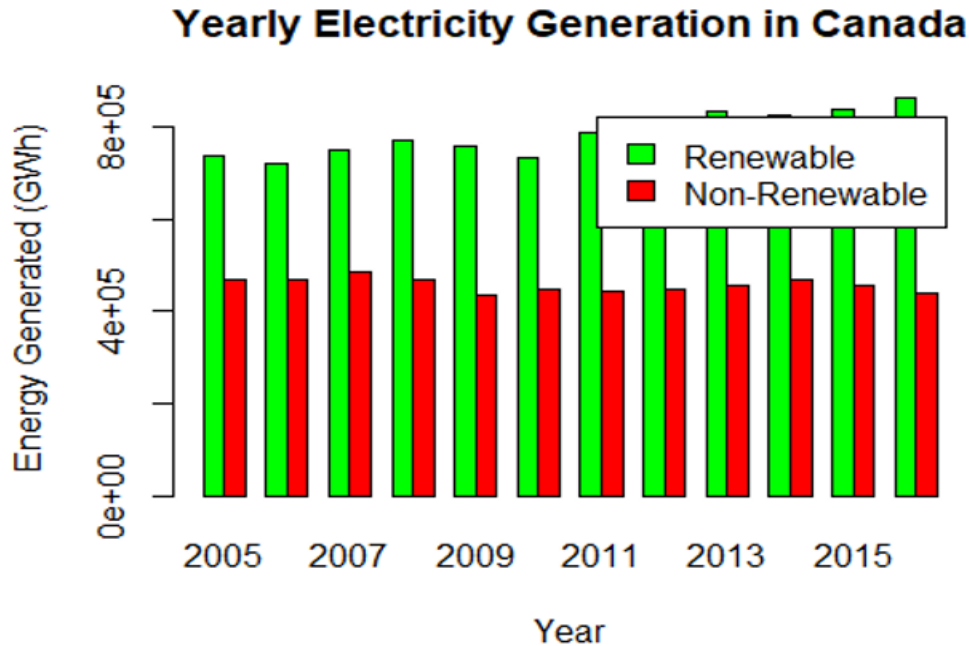


Figure 1. Bar plot of yearly renewable vs non-renewable generation.

3 METHODS

A combination of statistical techniques and visualizations was used to analyze differences between renewable and non-renewable electricity generation.

3.1 MAXIMUM LIKELIHOOD ESTIMATION (MLE)

MLE was used to estimate the mean and standard deviation of annual renewable and non-renewable energy generation, assuming a normal distribution. MLE was used because it provides reliable estimates of the mean and standard deviation with small sample sizes and allows both parameters to be estimated consistently under the assumption of normally distributed annual totals. This makes it well-suited for yearly electricity generation data, where only twelve observations are available for each energy category.

3.2 CONFIDENCE INTERVALS

Using the MLE-derived parameters, 95% confidence intervals were constructed for both the mean and standard deviation of each energy category. These intervals quantify uncertainty and help assess the stability of each energy source over time.

3.3 HYPOTHESIS TESTING

Two sample t tests were used to compare:

- renewable vs non renewable mean generation
- early period (2005–2010) vs late period (2011–2016)

These tests determine whether observed differences are statistically significant.

3.4 VISUALIZATION

Several charts were created to support interpretation, including:

- a time series scatter plot with trendlines
- a bar plot of yearly totals
- boxplots comparing distributions

These visualizations help illustrate the statistical findings and highlight meaningful trends.

4 KEY FINDINGS

4.1 POINT ESTIMATION

MLE results show that renewable energy production is both higher on average and more variable than non-renewable production. Renewable generation averaged 783,872.7 GWh with a standard deviation of 45,147.66 GWh, while non-renewable generation averaged 456,391.4 GWh with a standard deviation of 14,517.84 GWh. These values indicate that renewable energy not only dominates in total production but also fluctuates more from year to year.

4.2 TIME SERIES ANALYSIS

The time-series plot reveals a clear upward trend in renewable energy generation over the 12-year period. In contrast, non-renewable energy remains relatively stable, showing only minor fluctuations. Trendlines and horizontal MLE mean lines reinforce these patterns, highlighting the growing contribution of renewable sources to Canada's energy mix.

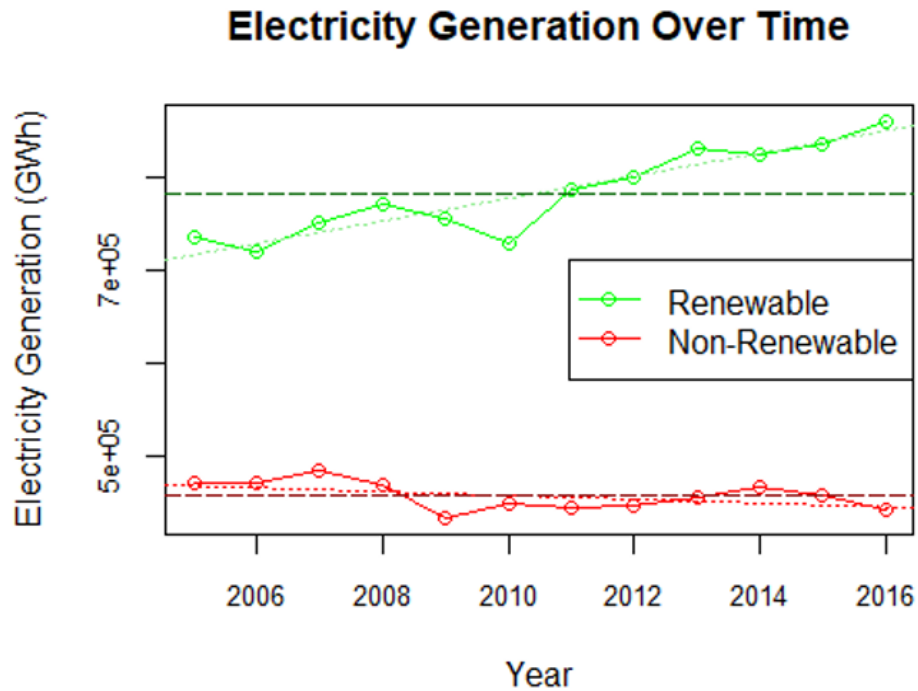


Figure 2. Time series plot with trendlines.

4.3 CONFIDENCE INTERVALS

Renewable Energy

- Mean CI: [758,328.4, 809,416.9]
- Standard deviation CI: [31,982.38, 76,655.22]

Non-Renewable Energy

- Mean CI: [448,177.3, 464,605.4]
- Standard deviation CI: [10,284.37, 24,649.53]

These intervals confirm that renewable energy has a wider confidence interval for both the mean and standard deviation, reflecting greater variability and uncertainty. Non-renewable energy shows narrower intervals, consistent with its more stable production levels.

4.4 HYPOTHESIS TESTING

4.4.1 Renewable vs Non-Renewable Means

The t-test comparing renewable and non-renewable means produced a p-value of $4.824e-12$, indicating a highly significant difference. Renewable energy is consistently higher and more variable across all years.

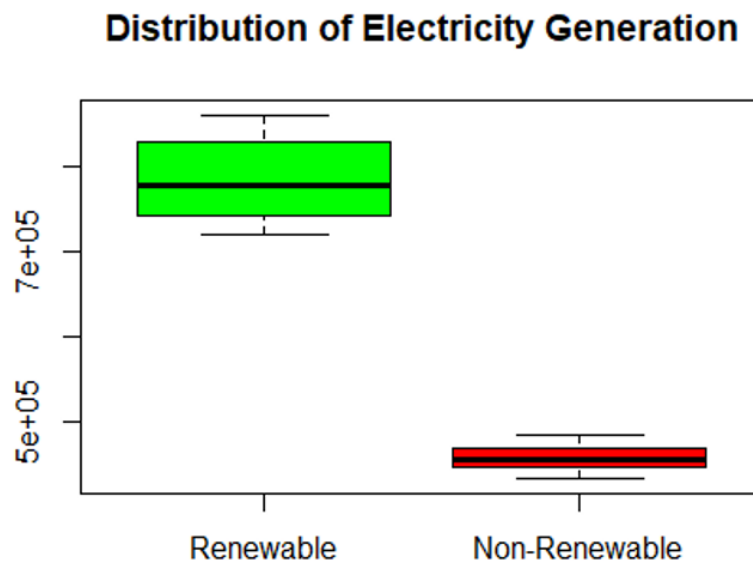


Figure 3. Boxplot comparing renewable vs non-renewable distributions

4.4.2 Period Comparison: 2005–2010 vs 2011–2016

For renewable energy, the p-value of 0.0001997 indicates a significant increase in the later period (2011–2016). Mean renewable generation rose from 743,948.6 GWh to 823,761.4 GWh, confirming substantial growth. For non-renewable energy, the p-value of 0.2465 indicates no significant change between periods, suggesting that production remained stable.

Distribution of Renewable Electricity Generation

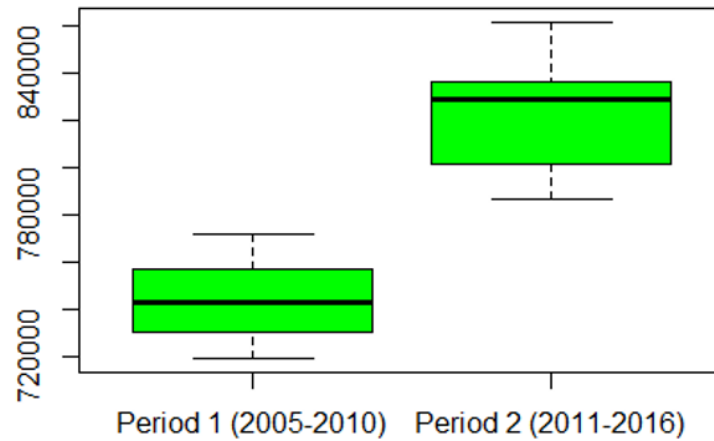


Figure 4. Boxplot comparing renewable energy from 2005-2010 vs 2011-2016

Distribution of Non-renewable Electricity Generation

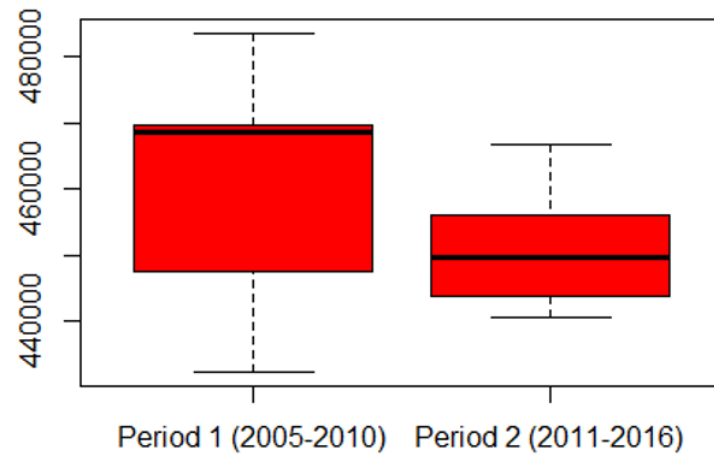


Figure 5. Boxplot comparing non-renewable energy from 2005-2010 vs 2011-2016

5 LIMITATIONS

Although the analysis provides meaningful insights, several limitations should be acknowledged. The dataset aggregates electricity generation at the national level, which obscures important provincial differences in energy production and policy. Renewable energy variability may also be influenced by weather patterns or hydrological conditions that are not

captured in the dataset. The statistical methods used assume independence between yearly observations and normality in the distribution of annual totals, assumptions that may not fully reflect the complexities of real-world energy systems. Additionally, the dataset ends in 2016, meaning that more recent developments in renewable technology and policy are not represented.

6 NEXT STEPS

Future work could expand this analysis in several ways:

- Incorporating data beyond 2016 to capture more recent renewable energy growth
- Conducting a provincial-level analysis to identify regional drivers of change
- Integrating weather or climate variables to explain year-to-year variability in renewable generation
- Applying time-series forecasting methods to project future production levels
- Evaluating the impact of major policy changes, such as carbon pricing or renewable incentives
- Comparing Canada's energy trends with those of other countries to contextualize national progress

7 REFERENCES

- Secretariat, T. B. of C., & Secretariat, T. B. of C. (n.d.). *Open Government Portal*. Open.canada.ca. <https://open.canada.ca/data/en/dataset/2cdf43fc-d4aa-4604-9f21-29777d955810/resource/b751e0dc-1a00-417a-9f04-b4777adb521f>